# How do social media algorithms shape the visibility of underrepresented communities in digital activism

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INFO 300: Research Methods
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June 5th, 2026

**Abstract**

Social media platforms have become central spaces for political discussion, public awareness, and digital activism. The visibility of online content is increasingly shaped by algorithmic recommendation systems that influence what users see, engage with, and share. This qualitative study examines how social media algorithms affect the visibility of activist content, particularly for underrepresented communities and movements that challenge dominant political or cultural narrative. Using digital ethnography and semi-structured interviews, this research explores the experiences of at least 15 college students and young adult social media users between the ages of 18 to 35 across TikTok, Instagram, and X. The research investigates participants' perceptions of algorithmic bias, censorship, shadow banning, and unequal visibility online. It is anticipated that participants will report modifying their posting behavior, language choice, and engagement strategies in response to perceived content suppression or reduced visibility. By examining user experience and engagement patterns, this study contributes to broader discussions about digital activism, algorithmic power, and freedom of expression in online spaces. Ultimately, the research seeks to better understand how platform algorithms shape civic discourse during periods of political tension, protest, and social change.

**Introduction**

Social media has become deeply integrated into everyday life, with billions of people using digital platforms to access information, communicate, and engage with current events. Yet, a question remains, how do these platforms choose what to inform us about, and how do they decide which issues are deemed important enough to show to a massive global population? Think back to your childhood or significant historic moments, such as a major presidential election, and consider how you first learned about them? Was it in a classroom, or from broadcast news, or did it get filtered through an algorithm on your social media feed?. Today, online networks have largely replaced traditional media as the primary source of public information, making the design of these platforms a major force in shaping public awareness and political mobilization.

As social media continues to replace traditional media as a major source of information, concerns about algorithmic bias and content visibility have become increasingly common. Activists and creators have raised concerns about censorship, shadow banning, and unequal visibility, particularly when discussing controversial political topics or issues connected to marginalized communities. Rather than acting as completely neutral public spaces, social media platforms may prioritize content that aligns with engagement goals, advertiser interests, or dominant cultural narratives while limiting the reach of other perspectives.

This research is important because social media now plays a significant role in shaping civic participation, political awareness, and public understanding of social issues. While existing studies have begun to explore algorithmic resistance, finding that activists often adapt their language and online behavior to navigate perceived censorship and maintain visibility (Elmimouni et al., 2025), there remains a gap in research examining the personal, lived experiences of college students. Much of the current research emphasizes platform structures or engagement metrics rather than how users personally interpret and navigate perceived suppression in their daily lives.

This study addresses that gap by examining how college students and young adults interpret and experience algorithmic visibility across social media platforms. Through digital

ethnography and semi-structured interviews, the project explores how users perceive amplification, marginalization, censorship, and unequal visibility online political discourse. Understanding these experiences is important because algorithmic systems increasingly influence whose voices are seen, shared, and discussed in public digital spaces.

**Literature Review/Prior Work**

Social media has become a key tool for activism, political participation, and community building in contemporary society. Platforms such as Instagram, TikTok, Facebook, and X enable users to organize movements, spread awareness, and challenge institutional systems (Cammaerts, 2015). Cammaerts (2015) notes that these platforms support faster cross-border communication and network building among activist groups.. Similarly, Cortés-Ramos et al. (2021), in a study of 58 high school students in Málaga, Spain, find that young people view social media as an accessible space for engaging with political and social issues. Together, this research suggests that social media has expanded opportunities for participation, particularly among younger users.

Although social media enables activism, visibility is increasingly shaped by algorithmic systems. Chon and Park (2019) argue that online activism depends on visibility and engagement, as content must reach audiences to generate impact. However, this visibility is largely controlled by recommendation and moderation systems. Schlosser et al. (2024) describe TikTok's recommendation system as "algorithmic doors" that can both amplify activist content and expose users to suppression or moderation. This indicates that algorithms do not simply organize content, but actively influence how activism circulates and which voices are made visible.

Recent studies and researchers have also examined how marginalized and underrepresented communities experience unequal visibility online. Uldam (2018) highlights the dual nature of visibility, noting that being seen can support "recognition" and identity formation, particularly in relation to issues such as class and identity politics, while also remaining dependent on platform-controlled systems like algorithmic ranking and moderation. Similarly, Sannon et al. (2023) emphasize that participation in online activism is especially important for underrepresented communities facing accessibility barriers, while also pointing to risks such as privacy concerns for marginalized groups engaging in advocacy content. Expanding on these challenges, Elmimouni et al. (2025) find that activists respond to censorship through a range of strategies, including self-censorship and adapting content in order to maintain visibility. Furthermore, these studies suggest that algorithmic systems not only shape what content is seen, but also influence how activists communicate and adapt their practices online.

Overall, existing research demonstrates that social media platforms play a major role in shaping activism, communication, and public discourse. Scholars generally agree that digital platforms increase opportunities for awareness, participation, and community building while also creating challenges related to algorithmic visibility, moderation, and unequal representation that disproportionately silence underrepresented communities. However, there remains a gap in research examining how college students and young adults personally experience and interpret algorithmic visibility in everyday social media use. As the primary demographic on these platforms, this sample directly represents the young adult population experiencing this daily bias. This study builds on existing research by focusing specifically on how users perceive amplification, suppression, and bias across platforms such as TikTok, Instagram, and X.

**Research Questions**

Therefore, the proposed study seeks to understand how social media algorithms shape digital activism by addressing the following research questions: How do social media algorithms influence the visibility, amplification, or suppression of activist content from underrepresented communities? How do users involved in or exposed to digital activism perceive algorithmic bias, censorship, shadow banning, and unequal visibility on platforms such as TikTok, Instagram, and X?

This study mainly focuses on the relationship between social media activism, algorithmic bias, and social impact on college students aged 18 to 24 who use social media as a space for political awareness and engagement. Specifically, it examines how different algorithms across three platforms may shape which activist voices are amplified, recommended, hidden or made less visible online. Since social media is now a major space for public awareness, political discussion, and organizing collective action, unequal algorithmic visibility can affect whose experiences and concerns become part of broader public conversation.

Additionally, the goal of this study is to better understand how algorithmic visibility affects the visibility of activist content created by or about underrepresented communities, especially as college students encounter, share, and interpret this content online. Through digital ethnography and semi-structured interviews, the study aims to identify patterns in engagement, censorship, amplification and marginalization. The broader outcome is to contribute to discussions about how social media platforms can become more transparent, fair and inclusive in the way activist content is shared, recommended and made visible across different channels. This connects directly to the project's focus on social media activism, algorithmic bias, and the impact these systems have on society.

**Research Design**

This proposed study will use a qualitative research design to understand how social media algorithms shape the visibility of digital activism, especially for underrepresented communities. A qualitative approach is the best fit because the study is not only trying to measure engagement numbers, but also understand people's experiences, perceptions, and interpretations of algorithmic bias, censorship, shadowing banning and unequal visibility online. Since the research questions focus on user experiences with algorithmic visibility, a qualitative approach allows us to better understand their perspectives, concerns, and examples of bias than a survey alone would.

The study will use two main methods: digital ethnography and semi-structured interviews. Digital ethnography will allow us to observe public social media activism posts on platforms including TikTok, Instagram, and X by looking at hashtags, captions, comments, likes, shares, reposts and patterns of engagement. This method is useful because the project focuses on how activism appears and circulates in real online spaces. Semi-structured interviews will allow us to understand the personal experiences of students and young adults who use or engage with social media activism, especially their perceptions of amplification, marginalization, censorship, and platform trust. These two methods work well together because digital ethnography shows visible patterns online, while interviews help explain how users interpret those patterns.

This research design is connected to an interpretivist epistemological standpoint because the study focuses on meaning, lived experience, and social interpretation. Instead of assuming that algorithmic bias can only be understood through platform metrics, this study recognizes that

users' experiences and perceptions are important forms of knowledge. The study also connects to a critical theoretical perspective because it examines how platform algorithms may reproduce unequal power structures by amplifying some voices while suppressing others. This matters because the project focuses on social media activism, algorithmic bias, and the broader impact on society, especially for marginalized or underrepresented communities. Although quantitative or experimental approaches could identify broad engagement patterns, they would not fully explain how users interpret suppression, amplification, or algorithmic bias. Because platform algorithms are not fully transparent, a qualitative design is more realistic and better suited for examining user experiences and public-facing patterns of visibility.

**Methods**

As described above, this study will use two qualitative methods: digital ethnography and semi-structured interviews. The first method will be digital ethnography. For this part of the study, we will observe, document and catalogue at least 50 viral or widely engaged political or activist posts from each platform: TikTok, Instagram, and X. This means the digital ethnography sample will include at least 150 total public posts posted between 2020 to 2025. This time period is useful because it includes major periods of online activism and increased public concern about platform moderation and algorithmic bias. We will focus on posts related to activism, underrepresented communities, algorithmic bias, censorship, and shadow banning. Search terms and hashtags may include phrases connected to social justice movements, platform suppression, and political activism. For each post, we will record information such as captions, hashtags, comments, likes, shares, reposts, and whether users mention reduced visibility or censorship. This method aligns with the research problem because it allows us to observe how activist content appears, circulates, and receives engagement in real social media spaces.

The second method will be semi-structured interviews. We will interview at least 15 college students and young adults aged 18 to 35 who use, post, share, or engage with social media activism. Each interview will last up to 30 minutes. Participants will be asked open-ended questions about their experiences with activist content, algorithmic visibility, censorship, shadow banning, platform trust, and whether they believe certain voices are amplified or suppressed online. Semi-structured interviews are appropriate because they give participants room to explain their experiences in their own words while still allowing the research team to ask consistent questions across broader themes such as platform use, experiences with activist content, perceptions of algorithmic bias or censorship, and trust in social media platforms. Participants will be offered a $20 reimbursement for their time from the allocated research budget.

The study population will include college students and young adults who engage with social media activism, especially on TikTok, Instagram, and X. The sampling strategy will be purposive sampling, because we want to recruit participants who have direct experience viewing, sharing, or creating activist content online. The sampling frame may include university students, student organizations, activist groups, and young adults who are active on social media. The sample population will likely include casual social media users as well as people who have posted or interacted with activism-related content. This approach fits the study because the goal is not to generalize to all social media users, but to understand detailed experiences and patterns related to algorithmic visibility.

For data collection, the digital ethnography portion will use a structured observation and coding sheet. We will document the platform, hashtag or search term used, type of activist content, visible engagement levels, user comments about suppression or amplification, and any

signs of censorship or reduced visibility. For the interview portion, participants will complete an informed consent form before the interview begins. With permission, interviews will be audio recorded, transcribed, and anonymized before analysis. If a participant does not give permission to be recorded, the research team will take detailed written notes during the interview instead. Interview responses and transcripts will be coded for themes such as amplification, marginalization, censorship, shadow banning, platform trust, and visibility challenges.

Together, these methods provide a strong way to answer the research questions. Digital ethnography helps identify visible patterns in activist content and engagement, while interviews help explain how users understand and experience those patterns. Using both methods also strengthens the study because we can compare what we observe online with what participants describe in interviews. This makes the findings more reliable and better connected to the broader issue of how algorithmic bias may affect social media activism and public visibility.

**Anticipated Results**

We anticipate finding that social media algorithms influence the visibility of activists' content by amplifying some voices while limiting the reach of others, particularly those from underrepresented communities. We expect participants to report experiences related to censorship, shadow banning, unequal visibility, and changing posting behaviors in response to perceived algorithmic bias. We also anticipate observing patterns in engagement that suggest certain topics, hashtags, or forms of activism receive greater visibility than others.

The collected data will be analyzed using thematic analysis to identify recurring themes across participants' responses and observed online content. Expected themes include amplification, marginalization, platform trust, visibility challenges, and strategies users employ to maintain engagement. Findings from the digital ethnography and interviews will be compared to identify areas of agreement and divergence.

To strengthen validity, the study will examine multiple social media platforms, activist topics, and participant perspectives. Triangulation will be used by comparing themes identified in the ethnographic observations with those emerging from participant interviews. Reliability will be supported through the use of standardized coding procedures, consistent interview questions, and clearly documented inclusion criteria. The use of multiple data sources will help ensure that the findings accurately reflect both observable online patterns and participant experiences.

These anticipated findings will help answer the research questions by identifying how users perceive algorithmic visibility and by revealing patterns that may indicate the amplification or suppression of activist content. Together, the findings will provide insight into how social media algorithms shape digital activism and influence the experiences of underrepresented communities online. The findings are expected to contribute to a greater understanding of how algorithmic systems shape digital activism and public discourse. They may also provide insight into the ways social media platforms influence representation, participation, and visibility in online spaces.

**Ethical Considerations**

This study raises several ethical considerations because it examines digital activism, algorithmic bias, and user experiences on social media platforms. One potential challenge is that participants may feel uncomfortable discussing experiences with censorship, shadow banning, discrimination, or political activism online. Some participants may worry that sharing these

experiences could negatively affect their online presence or expose them to criticism or harassment. In addition, the sensitive political and social topics often associated with digital activism may create risks if participant identities are revealed.

Since this study involves human subjects through semi-structured interviews and digital ethnography, protecting participant privacy and confidentiality is essential. Although social media posts are publicly available, users may not expect their content to be analyzed in a research setting. There is also a risk of misinterpreting engagement patterns or visibility metrics without access to the internal systems that determine algorithmic recommendations. To address these concerns, the study will avoid collecting private messages or personal identifying information, and all usernames, quotes, and participant responses will be anonymized in the final analysis.

Prior to data collection, the study proposal would be submitted to the university's Institutional Review Board (IRB) or equivalent ethics committee for review and approval. The ethics application would include the research purpose, recruitment procedures, consent materials, interview protocols, and plans for protecting participant data. The IRB review process helps ensure that the study follows established ethical guidelines and adequately protects participants from potential harm.

All participants will provide informed consent before participating in interviews. Consent forms will explain the purpose of the study, how collected data will be used, and participants' right to withdraw at any time without penalty. Interviews will only be recorded with participant permission, and all collected data will be stored securely. These procedures are intended to protect participant autonomy, confidentiality, and privacy while ensuring the research is conducted responsibly and ethically.

**Limitations**

Several limitations may affect the scope and interpretation of this study. First, researchers do not have access to the internal systems used by social media platforms, including visibility metrics, recommendation algorithms, impression logs, or moderation records. As a result, it is difficult to determine whether changes in engagement are directly caused by algorithmic suppression, platform policies, or other factors. This limitation makes it challenging to draw definitive conclusions about how algorithms influence the visibility of activist content.

Another limitation is the study population. Because the study focuses on a small qualitative sample of at least 15 university students and young adults, many participants may be casual social media users rather than content creators with large audiences. Their experiences may provide valuable insight into perceptions of algorithmic bias, but they may not fully reflect the experiences of professional activists, influencers, or individuals who rely heavily on social media for advocacy work. Additionally, the sample is geographically limited and may not represent the experiences of users from different regions, countries, age groups, or cultural backgrounds. Therefore, the findings should not be generalized to all social media users or activist communities.

To reduce the impact of these limitations, the study will document observable engagement metrics such as views, likes, comments, shares, and reposts when available. We will also compare engagement patterns across different types of content and examine both highly visible and low-visibility posts. By combining digital ethnography with interview data, the study can compare observable online patterns with participants’ lived experiences, providing a more comprehensive understanding of how algorithmic visibility may shape digital activism.

## Discussion

This study contributes to a growing body of research examining how social media algorithms shape digital activism and the visibility of underrepresented communities. As social media becomes an increasingly important space for political communication, community organizing, and public awareness, understanding how algorithmic systems influence visibility is critical. The findings may provide insight into how platform design affects whose voices are amplified, whose perspectives receive less attention, and how users navigate these systems in their everyday online experiences.

Beyond contributing to academic discussions about algorithmic bias, this research may have practical implications for activists, social media users, and platform developers. A better understanding of how underrepresented communities experience amplification, marginalization, and unequal visibility may encourage conversations about transparency, accountability, and fairness in algorithmic systems. The study may also highlight how users adapt their behavior in response to perceived censorship, shadow banning, or reduced reach.

The results of this study could be shared through academic presentations, research reports, or future publications focused on technology, communication, and social justice. While the study is limited in scope, it may serve as a foundation for future research involving larger and more diverse populations, additional social media platforms, and evolving algorithmic systems. Ultimately, this research seeks to contribute to ongoing conversations about creating more equitable and inclusive digital spaces for underrepresented communities and activists.

## Conclusion

As social media continues to play a major role in shaping public awareness and participation, questions about algorithmic visibility have become increasingly important. This proposed research contributes to ongoing discussions about how digital platforms influence communication, representation, and engagement in online spaces. By examining user experiences alongside observable online patterns, the study aims to provide a more nuanced understanding of the relationship between social media algorithms and digital activism. The findings may help inform future conversations about platform transparency, accountability, and equity while providing a foundation for continued research on the social impacts of algorithmic systems.

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